

EMPLOYMENT

Department of Biological Sciences, Purdue University
Irwin Tessman Assistant Professor

West Lafayette, Indiana
August 2025–Present

Channing Division of Network Medicine (CDNM)
Brigham and Women's Hospital, Harvard Medical School
Postdoctoral Research Fellow

Boston, Massachusetts
June 2021–May 2025

EDUCATION

University of Illinois Urbana-Champaign (UIUC)
Ph.D. in Physics, Advisor: Dr. Sergei Maslov

Urbana, Illinois
August 2014–May 2021

- *Dissertation title*: “Modeling for microbial communities with cross-feeding and predator-prey interactions”
- *Committee*: Sergei Maslov, Ido Goldenberg, James O'Dwyer, and Nigel Goldenfeld (Chair)

University of Science and Technology of China (USTC)
B.S. in Applied Physics, *summa cum laude*

Hefei, Anhui
September 2010–June 2014

RESEARCH INTERESTS

My research seamlessly blends computational biology, microbiology, ecology, biophysics, and medicine to devise computational methods for complex biological systems. I develop **mathematical models and machine learning methods that integrate multi-omics data to explore diet-microbe-metabolite-disease relationships**, revealing their dynamics and connection to human health.

Our research group aims to create computational approaches for translational research. Our long-term goal is to formulate novel computational methods to design dietary, prebiotic, probiotic, and phage therapies by targeting gut microbiomes.

SCHOLARSHIPS AND AWARDS

- Irwin Tessman termed Assistant Professorship 2025-2028
- National University of Singapore Presidential Young Professorship (PYP), ~1.5M USD, offered & declined 2025
- Fondazione Internazionale Menarini “Artificial Intelligence and Network Medicine: Path to Precision Medicine” Program Scholarship 2024
- The Les Houches School of Physics “Evolution of Diversity” Program Scholarship 2018
- Guo Moruo Scholarship (the highest honor in USTC) 2014
- UCLA Cross-disciplinary Scholars in Science and Technology (CSST) Summer Research Scholarship 2013
- China National Scholarship 2012
- XinDi Scholarship 2011

Published

- Sun Z, **Wang T**, Lasky-Su JA, Litonjua AA, Weiss ST, Chavarro JE, Liu YY. Intrapartum caesarean delivery and childhood BMI trajectories in relation to the infant gut microbiome in the VDAART prospective birth cohort. *eBioMedicine*. 2026 Jul 1;129. DOI: [10.1016/j.ebiom.2026.106347](https://doi.org/10.1016/j.ebiom.2026.106347)
- **Wang T**, George AB, Maslov S. Higher-order interactions in auxotroph communities enhance their resilience to resource fluctuations. *Cell Systems*. 2026 Mar 18;17(3). DOI: [10.1016/j.cels.2025.101491](https://doi.org/10.1016/j.cels.2025.101491) — highlighted in IGB Newsletter; EurekAlert!; Phys.org
- **Wang T**, Gori BM, Weiss ST, Menichetti G, Liu YY. Revealing interactions between microbes, metabolites, and dietary compounds using genome-scale analysis. *Microbiome*. 2026 Jan 30;14(1):74. DOI: [10.1186/s40168-025-02312-6](https://doi.org/10.1186/s40168-025-02312-6)
- **Wang T**, Holscher HD, Maslov S, Hu FB, Weiss ST, Liu YY. Predicting metabolite response to dietary intervention using deep learning. *Nature Communications*. 2025 Jan 18;16(1):815. DOI: [10.1038/s41467-025-56165-6](https://doi.org/10.1038/s41467-025-56165-6)
- **Wang T**, Fu Y, Shuai M, Zheng JS, Zhu L, Chan AT, Sun Q, Hu FB, Weiss ST, Liu YY. Microbiome-based correction for random errors in nutrient profiles derived from self-reported dietary assessments. *Nature Communications*. 2024 Oct 22;15(1):9112. DOI: [10.1038/s41467-024-53567-w](https://doi.org/10.1038/s41467-024-53567-w)
- Wu L, Wang XW, Tao Z, **Wang T**, Zuo W, Zeng Y, Liu YY, Dai L. Data-driven prediction of colonization outcomes for complex microbial communities. *Nature Communications*. 2024 Mar 16;15(1):2406. DOI: [10.1038/s41467-024-46766-y](https://doi.org/10.1038/s41467-024-46766-y) — highlighted in ScienMag; Bik's Picks
- **Wang T***, Li L*, Figeys D, Liu YY. Pairing metagenomics and metaproteomics to characterize ecological niches and metabolic essentiality of gut microbiomes. *ISME Communications*. 2024 Jan 1;4(1):ycae063. DOI: [10.1093/ismeco/ycae063](https://doi.org/10.1093/ismeco/ycae063)
- George AB, **Wang T**, Maslov S. Functional convergence in slow-growing microbial communities arises from thermodynamic constraints. *The ISME Journal*. 2023 Sep;17(9):1482-1494. DOI: [10.1038/s41396-023-01455-4](https://doi.org/10.1038/s41396-023-01455-4)
- Aparicio A, **Wang T**, Saavedra S, Liu YY. Feasibility in MacArthur's Consumer-Resource Model. *Theoretical Ecology*. 2023 Sep 1;16(3):225-238. DOI: [10.1007/s12080-023-00566-0](https://doi.org/10.1007/s12080-023-00566-0)
- Sun Z, Liu J, Zhang M, **Wang T**, Huang S, Weiss ST, Liu YY. Removal of false positives in metagenomics-based taxonomy profiling via targeting Type IIB restriction sites. *Nature Communications*. 2023 Sep 1;14(1):5321. DOI: [10.1038/s41467-023-41099-8](https://doi.org/10.1038/s41467-023-41099-8)
- Li L*, **Wang T***, Ning Z, Zhang X, Butcher J, Serrana JM, Simopoulos CM, Mayne J, Stintzi A, Mack DR, Liu YY, Figeys D. Revealing proteome-level functional redundancy in the human gut microbiome using ultra-deep metaproteomics. *Nature Communications*. 2023 Jun 10;14(1):3428. DOI: [10.1038/s41467-023-39149-2](https://doi.org/10.1038/s41467-023-39149-2)
- **Wang T**, Wang XW, Lee-Sarwar KA, Litonjua AA, Weiss ST, Sun Y, Maslov S, Liu YY. Predicting metabolomic profiles from microbial composition through neural ordinary differential equations. *Nature Machine Intelligence*. 2023 Mar;5(3):284-293. DOI: [10.1038/s42256-023-00627-3](https://doi.org/10.1038/s42256-023-00627-3) — highlighted in IGB Newsletter
- Wang XW, **Wang T**, Schaub DP, Chen C, Sun Z, Ke S, Hecker J, Maaser-Hecker A, Zeleznik OA, Zeleznik R, Litonjua AA, DeMeo DL, Lasky-Su J, Silverman EK, Liu YY, Weiss ST. Benchmarking omics-based prediction of asthma development in children. *Respiratory Research*. 2023 Feb 26;24(1):63. DOI: [10.1186/s12931-023-02368-8](https://doi.org/10.1186/s12931-023-02368-8)
- Ranoa DR, Holland RL, Alnaji FG, Green KJ, Wang L, Fredrickson RL, **Wang T**,..., Burke MD. Mitigation of SARS-CoV-2 Transmission at a Large Public University. *Nature Communications*. 2022 Jun 9;13(1):3207. DOI: [10.1038/s41467-022-30833-3](https://doi.org/10.1038/s41467-022-30833-3) — highlighted in EurekAlert!; MedicalXpress; Chicago Tribune; ScienMag
- Wang Z, Goyal A, Dubinkina V, George AB, **Wang T**, Fridman Y, Maslov S. Complementary resource preferences spontaneously emerge in diauxic microbial communities. *Nature Communications*. 2021 Nov 18;12(1):6661. DOI: [10.1038/s41467-021-27023-y](https://doi.org/10.1038/s41467-021-27023-y) — highlighted in EurekAlert!; Phys
- Tkachenko AV, Maslov S, **Wang T**, Elbana A, Wong GN, Goldenfeld N. Stochastic social behavior coupled to COVID-19 dynamics leads to waves, plateaus, and an endemic state. *Elife*. 2021 Nov 8;10:e68341. DOI: [10.7554/eLife.68341](https://doi.org/10.7554/eLife.68341) — highlighted in UIUC Physics news; BNL News
- Goyal A*, **Wang T***, Dubinkina V, Maslov S. Ecology-guided prediction of cross-feeding interactions in the human gut microbiome. *Nature Communications*. 2021 Feb 26;12(1):1335. DOI: [10.1038/s41467-021-21586-6](https://doi.org/10.1038/s41467-021-21586-6) — highlighted in EurekAlert!; Bik's Picks

- Pilosof S, Alcalá-Corona SA, **Wang T**, Kim T, Maslov S, Whitaker R, Pascual M. The network structure and eco-evolutionary dynamics of CRISPR-induced immune diversification. *Nature Ecology & Evolution*. 2020 Dec;4(12):1650-60. DOI: [10.1038/s41559-020-01312-z](https://doi.org/10.1038/s41559-020-01312-z) — highlighted in EurekAlert!; Phys
- Liao C, **Wang T**, Maslov S, Xavier JB. Modeling microbial cross-feeding at intermediate scale portrays community dynamics and species coexistence. *PLoS Computational Biology*. 2020 Aug 18;16(8):e1008135. DOI: [10.1371/journal.pcbi.1008135](https://doi.org/10.1371/journal.pcbi.1008135)
- Ping D*, **Wang T***, Fraebel DT, Maslov S, Sneppen K, Kuehn S. Hitchhiking, collapse, and contingency in phage infections of migrating bacterial populations. *The ISME Journal*. 2020 Aug;14(8):2007-2018. DOI: [10.1038/s41396-020-0664-9](https://doi.org/10.1038/s41396-020-0664-9) — highlighted in Nature Research Microbiology Community; Bik's Picks
- **Wang T***, Goyal A*, Dubinkina V, Maslov S. Evidence for a multi-level trophic organization of the human gut microbiome. *PLoS Computational Biology*. 2019 Dec 19;15(12):e1007524. DOI: [10.1371/journal.pcbi.1007524](https://doi.org/10.1371/journal.pcbi.1007524) — highlighted in Phys; ScienceDaily; EurekAlert!; Bik's Picks

Review paper

- Wang XW, **Wang T**, Liu YY. Artificial intelligence for microbiology and microbiome research. *Cell Systems*. 2026 Feb 18;17(2). DOI: [10.1016/j.cels.2026.101531](https://doi.org/10.1016/j.cels.2026.101531)

Preprints Under Review

- Yang Y, **Wang T**, Huang D, Wang XW, Weiss ST, Liu YY. Deep learning transforms phage-host interaction discovery from metagenomic data. *Under Review and Revision at Nature Communications*. DOI: [10.1101/2025.05.26.656232](https://doi.org/10.1101/2025.05.26.656232)
- Zhou R, Shen X, Mina TH, Agrawal KR, Low DY, Kang JX, Teo JJ, How SH, Lam BC, **Wang T**, Chong CW, Sung JJY, Nagarajan N, Ali Y, Chambers JC, Wong SH. Distinct gut microbiome signatures in ethnically diverse populations within a shared urban asian geography. *Under Review and Revision at Gut Microbes*. DOI: [10.64898/2026.02.06.26345736](https://doi.org/10.64898/2026.02.06.26345736)
- Wei Y, Elgart V, **Wang T**, Loscalzo J, Wang XW. Modeling microbiome perturbations reveals ecological control of host metabolite profiles. *Under Review at Communications Biology*.

TEACHING EXPERIENCE

- Mentor: Research PhD Thesis, BIOL 69900, Purdue University, Summer 2026–Present
- Mentor: Undergraduate Biology Research, BIOL 29400/49400, Purdue University, Spring 2026–Present
- Lecturer: “Food Polysaccharides”. Food Chemistry, FOOD-SCI 541, University of Massachusetts Amherst, March 2025
- Invited Lecturer: “Regression Models – predicting metabolomic profiles of microbial communities as an example”. National University of Singapore, February 2025
- Lecturer: “Electric Fields”. General Physics II, PHYS 114, Wake Forest University, January 2025
- Invited Lecturer: “Application of Machine-Learning Methods in Biology and Medicine”. Summer School hosted by the School of Medicine, Peking University, July 2024
- Invited Lecturer: “Application of Machine-Learning Methods in Biology and Medicine”. Summer School hosted by the School of Medicine, Peking University, July 2023
- Guest Lecturer: “Regression Models – predicting metabolomic profiles of microbial communities as an example”. Microbiome Data Science, BIO/MIC 494/598, Arizona State University, March 2023
- Teaching Assistant: Statistical Physics, UIUC PHYS 504, Spring 2017
- Teaching Assistant: Relativity & Math Applications, UIUC PHYS 225, Fall 2016
- Teaching Assistant: Classical Mechanics II, UIUC PHYS 326, Spring 2016
- Teaching Assistant: Classical Mechanics II, UIUC PHYS 326, Fall 2015
- Teaching Assistant: Quantum Mechanics II, UIUC PHYS 581, Spring 2015
- Teaching Assistant: Classical Mechanics II, UIUC PHYS 326, Fall 2014

MENTORING

- Qiwen Chen, PhD student, Spring 2026–Present
- Fei Wang, PhD student, Fall 2025–Present
- Alex Wu, undergraduate student, Spring 2026–Present
- Jocelyn Yang, undergraduate student, Spring 2026–Present
- Pranay Shah, undergraduate student, Spring 2026–Present
- Seohyun Kang, undergraduate student, Spring 2026–Present

ALUMNI

- Md. Imran Hasan, rotation PhD student, Spring 2026
- Damrongrat Siriwan, rotation PhD student, Fall 2025

INVITED TALKS

- Deep learning for microbiome research and precision nutrition. Virtual oral presentation at the Institute for Environmental Genomics, University of Oklahoma, March 2026
- Computational methods for microbiome research and precision nutrition. Oral presentation at Purdue Applied Microbiome Sciences (PAMS), Purdue University, West Lafayette, IN, February 2026
- Computational methods for microbiome research and precision nutrition. Oral presentation at the Center for Computational Biology and Bioinformatics (CCBB), Indiana University School of Medicine, Indianapolis, IN, February 2026
- Higher-Order Interactions in Auxotroph Communities Enhance Their Resilience to Resource Fluctuations. Oral presentation for the EcoLunch Seminar Series, Department of Biological Sciences, West Lafayette, IN, September 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Center for Computational Life Sciences, Cleveland Clinic, Cleveland, OH, April 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Department of Biomedical Engineering, University of Connecticut, Storrs, CT, April 2025
- Computational methods for microbiome research and precision nutrition. Oral presentation at the Department of Medicine and Life Sciences, Universitat Pompeu Fabra, Barcelona, Spain, April 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Department of Food Science, University of Massachusetts Amherst, Amherst, MA, March 2025
- Computational methods for microbiome research and precision nutrition. Oral presentation at the Department of Biological Sciences, Purdue University, West Lafayette, IN, February 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Department of Nutritional Sciences, University of Oklahoma Health Sciences Center, Oklahoma City, OK, February 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Department of Biology, University of Texas at Arlington, Arlington, TX, February 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Synthetic Biology for Clinical and Technological Innovation (SynCTI), National University of Singapore, Singapore, February 2025
- Computational methods for microbiome research and precision nutrition. Virtual oral presentation at the Benioff Center for Microbiome Medicine, University of California, January 2025
- Quantitative methods for microbiome research and precision nutrition. Oral presentation at the Department of Physics, Wake Forest University, Winston-Salem, NC, January 2025
- Computational methods for microbiome research and precision nutrition. Virtual oral presentation at the Department of Computer Science and Engineering, University of Texas at Arlington, January 2025

- Predicting metabolic response to dietary intervention using deep learning. Virtual oral presentation at AI in Biomedicine, Keystone Symposia, May 2024
- Machine learning for precision nutrition. Virtual oral presentation at the Bio-Data Club Seminar, Moffitt Cancer Center, Tampa, FL, January 2024
- Machine learning for precision nutrition. Oral presentation at Circulating Metabolic Intermediates as Fuels and Signals Conference, Keystone Symposia, Salt Lake City, UT, October 2023
- Machine learning for precision nutrition. Oral presentation at the MIT Center for the Physics of Living Systems, Department of Physics, Massachusetts Institute of Technology, Boston, MA, April 2023
- Pairing metagenomics and metaproteomics to pinpoint ecological niches and metabolic essentiality of microbial communities. Oral virtual presentation at the International Workshop on Soft Matter and Biophysics theories, Institute of Theoretical Physics, Chinese Academy of Sciences, November 2022
- Pairing metagenomics and metaproteomics to pinpoint ecological niches and metabolic essentiality of microbial communities. Oral presentation at the MIT Center for the Physics of Living Systems, Department of Physics, Massachusetts Institute of Technology, Boston, MA, May 2022
- Pairing metagenomics and metaproteomics to pinpoint ecological niches and metabolic essentiality of microbial communities. Oral presentation at the School of Pharmaceutical Sciences, University of Ottawa, Ottawa, Canada, May 2022
- Predicting metabolomic profiles from microbial compositions through neural ordinary differential equations. Oral presentation at the Center for Complex Network Research, Northeastern University, Boston, MA, March 2022

CONTRIBUTED PRESENTATIONS

- Revealing the interactions between microbes, metabolites, and dietary compounds based on microbial genomes. Oral presentation at Microbial Communities, APS March Meeting, Los Angeles, CA, March 2025
- Machine learning for precision nutrition. Oral presentation at Cold Spring Harbor Laboratory Microbiome Meeting, Cold Spring Harbor, NY, October 2024
- Predicting metabolic response to dietary intervention using deep learning. Oral presentation at Microbial Communities, APS March Meeting 2024, Minneapolis, MN, March 2024
- Predicting metabolic response to dietary intervention using deep learning. Poster presentation at Microbial Ecology and Evolution Hub-based Conference 2024, Boston College, Boston, MA, January 2024
- Machine learning for precision nutrition. Poster presentation at IAIFI Summer Workshop, Northeastern University, Boston, MA, August 2023
- Microbiome-based correction of dietary assessment. Oral presentation at Health Professionals Follow-up Study meeting, Harvard Medical School, Boston, MA, May 2023
- Predicting metabolomic profiles from microbial compositions through neural ordinary differential equations. Poster presentation at MIT Microbiome Symposium, Massachusetts Institute of Technology, Boston, MA, April 2023
- Predicting metabolomic profiles from microbial composition through neural ordinary differential equations. Oral presentation at Microbial Communities, APS March Meeting 2023, Las Vegas, NV, March 2023
- Pinpointing ecological niches and metabolic essentiality of microbial communities using both metagenomics and metaproteomics. Oral presentation at Ecological and Evolutionary Biology, APS March Meeting 2022, Chicago, IL, March 2022
- Predicting metabolomic profiles from microbial composition through neural ordinary differential equations. Oral presentation at Channing Microbiome Meeting, Harvard Medical School, Boston, MA, February 2022
- Hitchhiking, collapse, and contingency in phage infections of migrating bacterial populations. Poster presentation at Biocomplexity Theme Review, Carl R. Woese Institute for Genomic Biology, Urbana, IL, October 2019
- CRISPR-induced Red Queen dynamics in the phage-microbial system. Poster presentation at Biocomplexity Theme Review, Carl R. Woese Institute for Genomic Biology, Urbana, IL, October 2019
- Evidence for a multi-level trophic organization of the human gut microbiome. Poster presentation at IGB Fellows Symposium, Carl R. Woese Institute for Genomic Biology, Urbana, IL, April 2019

- Thermodynamic constraints on cross-feeding in bacterial population. Oral presentation at Ecological and Evolutionary Biology, APS March Meeting 2019, Boston, MA, March 2019
- Hitchhiking, collapse, and contingency in phage infections of migrating bacterial populations. Poster presentation at the “Evolution of Diversity” Program, Les Houches Physics School, Les Houches, February 2018

PEER REVIEW

- Scientific Journals: Science, Cell, Nature Communications, PNAS, Trends in Microbiology, Microbiome, Genome Biology, eLife, PloS Computational Biology, Methods in Ecology and Evolution, Biophysical Journal, Microbiology Spectrum, Molecular Neurobiology, Scientific Data, Nutrients, Frontiers in Immunology, Computational and Structural Biotechnology Journal, PloS One, BMC Medicine, BMC Microbiology, peerJ
- Grants: UKRI (United Kingdom Research and Innovation)

SERVICES

- Official Nominator, VinFuture Prize, 2026–Present
- Graduate Student Admission Committee, Department of Biological Sciences, 2025–2026

MEMBERSHIPS

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|---|--------------|
| • Member, American Society for Microbiology (ASM) | 2026–Present |
| • Member, International Association for Intelligent Biology and Medicine (IAIBM) | 2026–Present |
| • Member, Eli Lilly and Company and Purdue University Research Alliance Center (LPRC) | 2026–Present |
| • Member, American Physical Society (APS) | 2017–Present |